



A SYSTEMATIC COMPARATIVE STUDY OF THREE NATURAL O/W EMULSIFIERS AND THEIR APPLICATION IN FINISHED PRODUCTS

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LM-71 Cosmetic Industrial Science



LabAnalysis Life Science

WHAT WE'LL COVER TODAY

01

Research Context

Thesis objectives

02

The Emulsifiers

Three natural O/W systems under investigation

03

Methodology

7 experimental series × 3 emulsifier systems

04

Key Results

Stability, rheology, sensory & microscopy findings

05

The Products

BB Cream · Milky Toner · Anti-Age Body Cream

06

Conclusions

Emulsifier-to-product match & outcomes

THESIS OBJECTIVE

Milan-based contract laboratory · Internship host company

Thesis Objective

1. Systematic comparative study

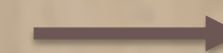
of three natural emulsifiers across 7 experimental series

2. Translate screening results

into 3 finished cosmetic products



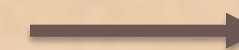
*Sorbitan Laurate /
Polyglyceryl-4 Laurate /
Dilauryl Citrate*



*3-component ester
blend
Non-ionic emulsifier*



*C14-22 Alcohols /
C12-20 Alkyl Glucoside*



*Alkyl polyglucoside
(APG) + fatty alcohol*



*Sodium Starch
Octenylsuccinate*



*Modified starch –
Pickering Emulsion*

EXPERIMENTAL DESIGN

7 independent series x 3 emulsifiers

Esters mix = 2%
APG = 2%
Modified Starch = 5%

01

Baseline Stability

Suggested emulsifier concentration
· 20% mixed oil · $\pm$ stabilisers

05

Rheology Modifiers

3 polymers (Xanthan Gum · Hydroxyethylcellulose · Polyacrylate)

02

Oil Phase Concentration

10% vs 30% oil load · stabilized system

06

Preparation Method

Modified Direct Emulsification vs. Lin Method

03

Emollient Category

5 single oils
· sub series A = without stabilisers
· sub series B = with stabilisers

07

Pigmented Emulsion

Inorganic pigments

04

Emulsifier Concentration Variation

Doubled or halved concentration · $\pm$ stabilisers

- Trygliceride
- Fatty Acid Ester
- Diester
- Branched Fatty Acid
- Hydrocarbon

Characterisation at t_0 , t_{24} + long term stability:

pH · Viscosity (2.5/5/10 rpm) · Centrifuge stability · Optical microscopy
+ Sensory evaluation across all three systems

KEY FINDINGS

RESULTS OVERVIEW

ESTERS MIX

- **Binary stabiliser** (XG + CA) essential for the stability
- **Best oil compatibility**: Coco-Caprylate/Caprates
- **Optimal oil range**: 10–20%; coalescence at 30%
- Good pigment compatibility → selected for **BB Cream**
- **Process**: 60°C is essential to preserve emulsifier activity

APG

- Intrinsic lamellar gel-network; high self-structuring
- **Better stability at 30%** than 10% oil load
- **Excluded from pigment series** (phase separation at t_{24h})
- Rich, structured cream texture → selected for **Body Cream**
- **Process**: 75°C; complete melting of all lipid components required

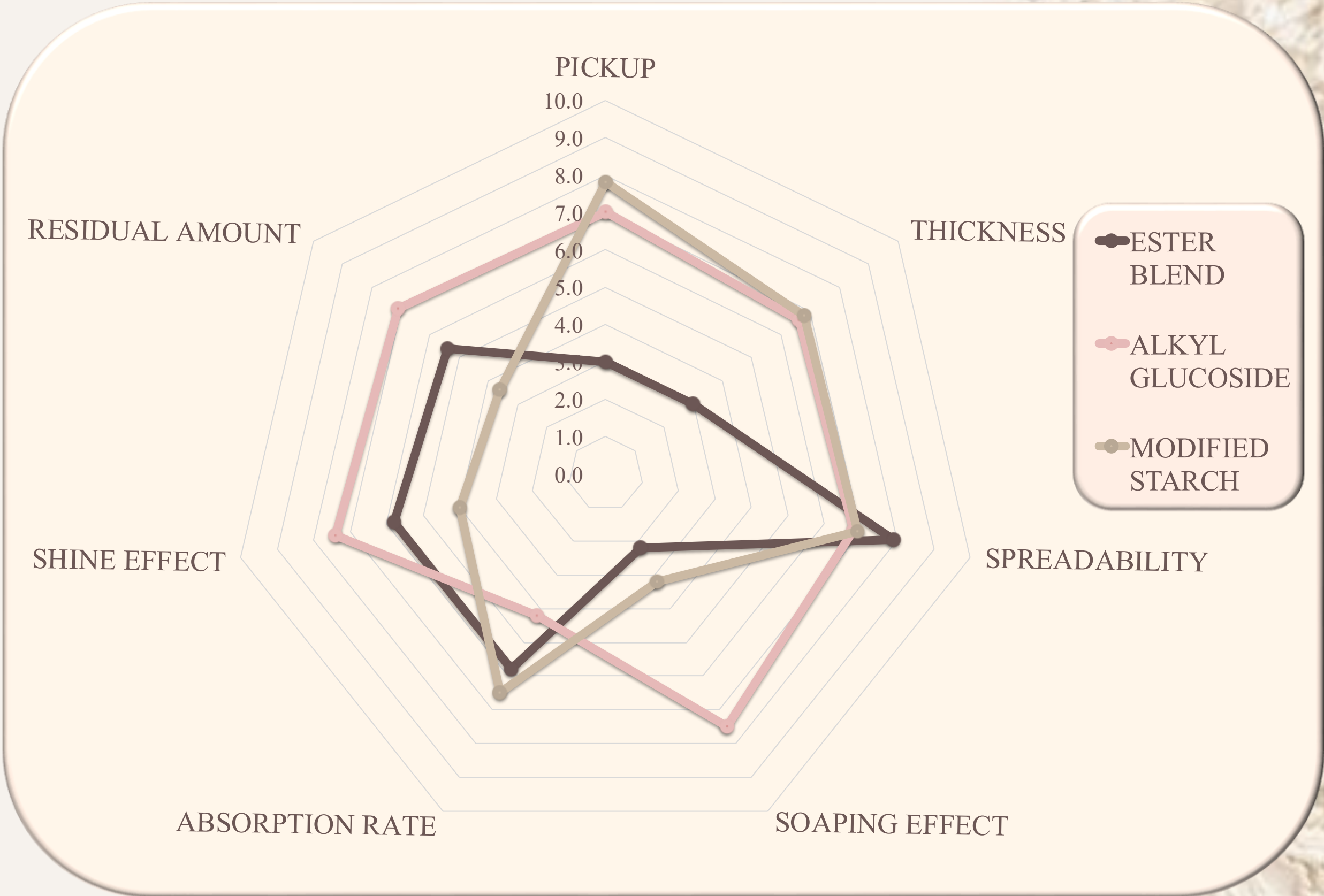
MOD. STARCH

- **Pickering emulsion** structure
- Surfactant free emulsifier
- **Needs 2,5/5% use level**; dispersed in aqueous phase
- **Stabilizer polymers** are requested
- Light milky texture at moderate oil load → selected for **Milky Toner**

SENSORY ANALYSIS

RESULTS OVERVIEW

Emulsifier Fingerprints — Panel n = 5, scale 0–10



ESTER BLEND

Light, silky texture. Highest spreadability (7.9), lowest soaping effect (2.2).

ALKYL GLUCOSIDE

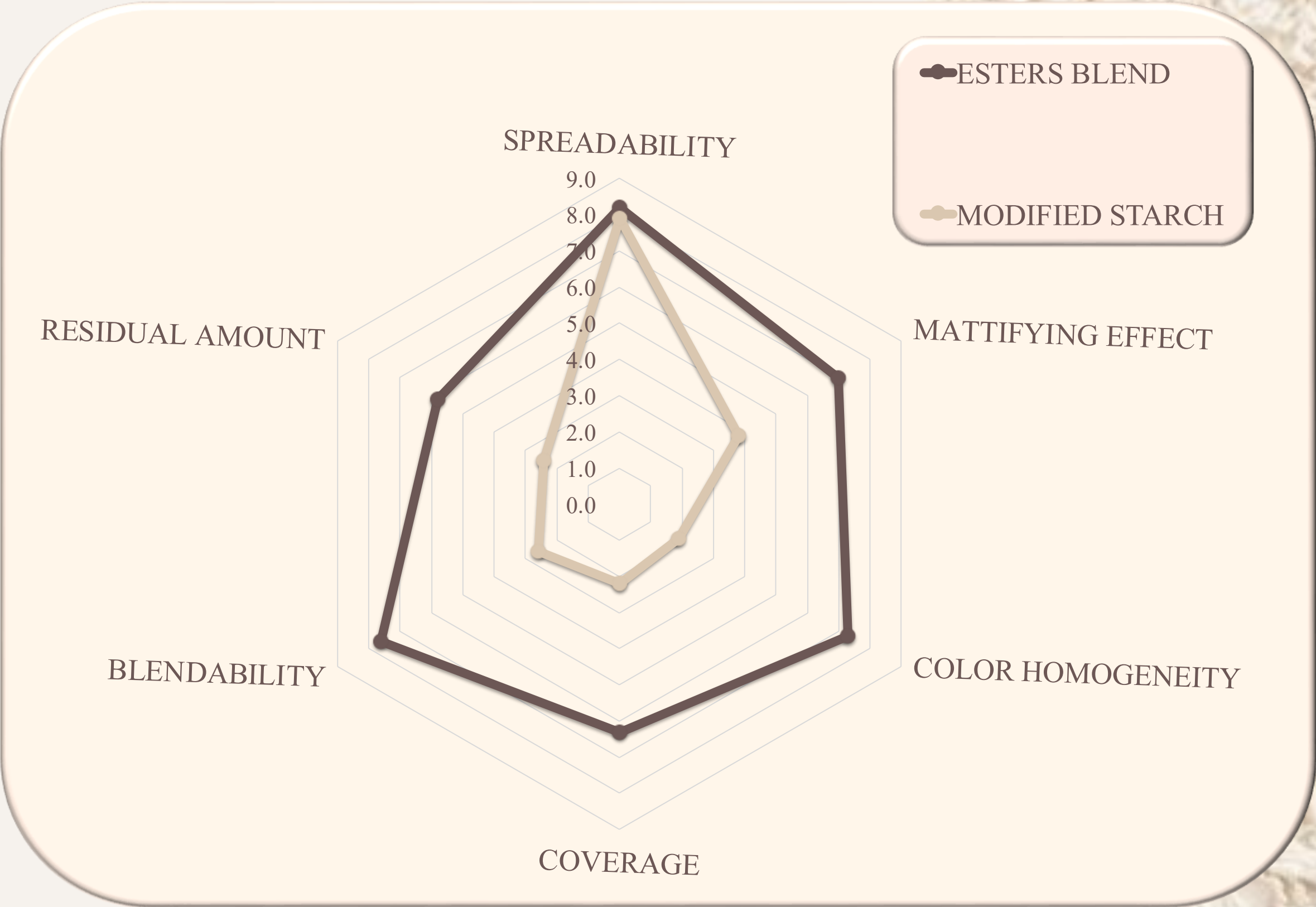
Rich, occlusive film. Highest shine (7.4) and residual amount (7.1); distinctive soaping effect (7.5).

MODIFIED STARCH

Balanced pick-up (7.8) and fastest absorption (6.5). Lowest residual (3.6) — clean, powdery finish.

SENS. ANALYSIS ON COLOURED EM.

Pigmented Emulsions — Panel n = 5, scale 0–10



ESTER BLEND

Outperforms in all 6 colour-specific attributes. Superior mattifying (7.0 vs 3.8), colour homogeneity (7.3 vs 1.9), and blendability (7.6 vs 2.6).

ALKYL GLUCOSIDE

It is not present because it is unstable.

MODIFIED STARCH

Rapid absorption (low residual = 2.4) limits pigment retention on skin — unsuitable for colour cosmetics

PRODUCTS DEVELOPMENT



BB CREAM
Anti-Pollution



BODY CREAM
Anti-Age



MILKY TONER
Hydrating

BB CREAM ANTI-POLLUTION

MAKE UP

Emulsifier: Sorbitan Laurate / Polyglyceryl-4 Laurate / Dilauryl Citrate @ 4%

Formulation

Phase	INCI Name	Conc. Range (% w/w)
A	Aqua, Xanthan Gum	65–75
A1	Chelating Agent, Glycerin, Ectoin	1–5
B	Caprylic/Capric Triglyceride, Coco-Caprylate/Caprate, Hydrogenated Polydecene, Cetearyl Alcohol, Sorbitan Laurate, Polyglyceryl-4 Laurate, Dilauryl Citrate	20–30
B1	Uncoated Pigment	3–5
C	Ammonium Acryloyldimethyltaurate/VP Copolymer	< 1
D	Active Ingredients	< 5
E	Parfum	< 0.5
F	Preservatives	0.5–1.5

pH: 6.7 / 6.8 · **Viscosity:** 8,960 / 19,520 mPa·s
Centrifuge: Pigment sedimentation; no phase separation
Microscopy: Uniform, 2–5 µm droplets

Anti-Pollution Active System

Ectoin

- Membrane & protein stabilisation;
- Stabilises lipid bilayer fluidity under oxidative stress

Lespedeza Capitata

- Circadian clock protection
- Mitigates AhR-mediated disruption by pollution & blue light;
- Flavonoid antioxidant activity

Camellia Japonica

- Direct radical scavenging (O_2^- , $\cdot OH$, 1O_2);
- MMP-1 inhibition in UVA-exposed fibroblasts;



BB CREAM ANTI-POLLUTION

MAKE UP

Emulsifier: Sorbitan Laurate / Polyglyceryl-4 Laurate / Dilauryl Citrate @ 4%

Formulation

Phase	INCI Name	Conc. Range (% w/w)
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B1	Uncoated Pigment	
C	Ammonium Acryloyldimethyltaurate/VP Copolymer	
D	Active Ingredients	< 5
E	Parfum	< 0.5
F	Preservatives	0.5–1.5

Examples of coating tried:

- Aluminum Hydroxide
- Silane Coated
- Sodium Lauroyl Glutamate + Lysine
- Phytic Acid
- Sodium Glycerophosphate

Anti-Pollution Active System

Ectoin

- Membrane & protein stabilisation;
- Stabilises lipid bilayer fluidity under oxidative stress

Lespedeza Capitata

- Circadian clock protection
- Prevents circadian clock disruption by pollution & blue light;

(O₂⁻, ·OH, ¹O₂);
α-exposed fibroblasts;



pH: 6.7 / 6.8 · **Viscosity:** 8,960 / 19,520 mPa·s
Centrifuge: Pigment sedimentation; no phase separation
Microscopy: Uniform, 2–5 µm droplets



BODY CREAM ANTI-AGE

BODY CARE

Emulsifier: C14-22 Alcohols / C12-20 Alkyl Glucoside @ 4%



Anti-Age Active System

Matrixyl → Matrikine KTTKTS

- Simultaneous stimulation of 6 ECM components: collagen I, III, IV; fibronectin; hyaluronic acid; laminin-5.
- Hydroxypropyl cyclodextrin delivery system

Niacinamide

- NAD⁺/NADP⁺ precursor;
- Stimulates ceramide synthesis;
- Inhibits melanosome transfer

Allantoin

- Keratolytic: promotes controlled desquamation; fibroblast & keratinocyte proliferation;
- Anti-inflammatory activity;
- Skin-conditioning on surface keratin

Formulation

Phase	INCI Name	Conc. Range (% w/w)
A	Aqua, Xanthan Gum	35–45
A1	Hydroxyethylcellulose	< 1
A2	Glycerin, Sodium Hyaluronate	5–10
A3	Chelating Agent, Active Ingredients, Preservative	1–5
B	Caprylic/Capric Triglyceride, Coco-Caprylate/Caprate, C13-15 Alkane, Butyrospermum Parkii Butter, Hydrogenated Ethylhexyl Olivatate, Hydrogenated Olive Oil Unsaponifiables, Caprylic/Capric/Myristic/Stearic Triglyceride, C14-22 Alcohols, C12-20 Alkyl Glucoside, Cetearyl Alcohol	30–40
C	Aqua	10–20
D	Preservatives and Antioxidant	0.5–1.5
E	Active Ingredients	< 5
F	Gold Pigment	<1

pH: 6.3 / 6.3 · **Viscosity:** 98,000 / 91,600 mPa·s
Centrifuge: Stable; light precipitate of gold pigment
Microscopy: Coarse but uniform, 2–20 µm droplets

MILKY TONER

K-BEAUTY

Emulsifier: Sodium Starch Octenylsuccinate (Pickering-like) @ 2.5%

Formulation

Phase	INCI Name	Conc. Range (% w/w)
A	Aqua, Xanthan Gum Crosspolymer-2	75–85
A1	Glycerin, Butylene Glycol, Sodium Hyaluronate	1–5
A2	Texture Modifier	1–5
A3	Chelating Agent, Preservative	<1
B	Coco-Caprylate/Caprate, C13-15 Alkane	5–10
C	Aqua, Active Ingredients	1–5
D	Preservatives	0.5–1.5
E	Ceramide Complex	<1

pH: 6.2 / 6.2 · **Viscosity:** 15,160 / 18,920 mPa·s
Centrifuge: Slight phase separation observed
Microscopy: Fine, 2–10 µm

Barrier & Microbiome Active System

Sodium Hyaluronate

- Water-binding capability reduces TEWL;
- High MW = reinforces natural moisturizing factors (NMF)

Centella Asiatica

- NF-κB suppression, and collagen stimulation;
- Soothing for post-cleansing reactive skin

Lactobacillus Ferment

- Postbiotic metabolites;
- Restores microbiome balance disrupted by surfactant cleansing

Ceramide Complex

- CER NP + AP + EOP
- Reconstitutes lamellar bilayer



MARKET POSITIONING

SKINIFICATION

BB CREAM & BODY CREAM

skincare actives and textures into make-up and body care

BB Cream:

Serum-weight texture anti-pollution for make-up with verified skin benefits

Body Cream:

Anti-age actives position a body cream with the skinification shift in premium body care



MARKET POSITIONING

CONCLUSION

MILKY TONER

K-BEAUTY

- **K-beauty ritual logic:** multi-step layering philosophy in which a low-viscosity,
- The microbiome-centred active system
- **Dispensing ritual:** the controlled pour that reinforces a meditative application gesture

- Unique performance of each emulsifier
- Validated product–emulsifier matching
- Three fully developed cosmetic products



THANK YOU FOR
YOUR ATTENTION

